

Restore Datafile to New Location (ASM to File System)

Step-by-Step RMAN & SQL*Plus Procedure with Verification Outputs

1. Check Initial Tablespace and Datafile Details

Verify the current file location and status of datafile 5 residing on ASM (+DATA).

```
SQL> SELECT FILE_ID, TABLESPACE_NAME, FILE_NAME, STATUS, AUTOEXTENSIBLE, MAXBYTES/
1024/1024/1024 AS "MAX_SIZE(GB)" FROM DBA_DATA_FILES WHERE FILE_ID=5;
```

FILE_ID	TABLESPACE_NAME	FILE_NAME	STATUS	AUT	MAX_SIZE(GB)
5	SHA	+DATA/PROD/sha01.dbf	AVAILABLE	YES	31.9999847

2. Take RMAN Backup of Datafile 5

Perform an RMAN backup of the target datafile prior to restoration.

```
RMAN> backup datafile 5;
```

```
Starting backup at 07-SEP-2026 13:55:30
using target database control file instead of recovery catalog
allocated channel: ORA_DISK_1
channel ORA_DISK_1: SID=126 device type=DISK
channel ORA_DISK_1: starting full datafile backup set
channel ORA_DISK_1: specifying datafile(s) in backup set
input datafile file number=000005 name=+DATA/PROD/sha01.dbf
channel ORA_DISK_1: starting piece 1 at 07-SEP-2026 13:55:33
channel ORA_DISK_1: finished piece 1 at 07-SEP-2026 13:55:34
piece handle=+FRA/PROD/BACKUPSET/2026_09_07/nnndf0_tag20260907t135532_0.268.1243346133
tag=TAG20260907T135532 comment=NONE
channel ORA_DISK_1: backup set complete, elapsed time: 00:00:01
Finished backup at 07-SEP-2026 13:55:34
```

```
Starting Control File and SPFILE Autobackup at 07-SEP-2026 13:55:34
piece handle=+FRA/PROD/AUTOBACKUP/2026_09_07/s_1243346134.269.1243346135 comment=NONE
Finished Control File and SPFILE Autobackup at 07-SEP-2026 13:55:37
```

3. Check Tablespace Status Before Taking Offline

```
SQL> SELECT FILE_ID, TABLESPACE_NAME, FILE_NAME, STATUS, AUTOEXTENSIBLE, MAXBYTES/
1024/1024/1024 AS "MAX_SIZE(GB)", ONLINE_STATUS FROM DBA_DATA_FILES WHERE FILE_ID=5;
```

FILE_ID	TABLESPACE_NAME	FILE_NAME	STATUS	AUT	MAX_SIZE(GB)	ONLINE_
5	SHA	+DATA/PROD/sha01.dbf	AVAILABLE	YES	31.9999847	ONLINE

4. Take Tablespace Offline

Take the tablespace offline immediately. Note that the online status must become RECOVER.

```
SQL> ALTER TABLESPACE SHA OFFLINE IMMEDIATE;
Tablespace altered.
```

```
SQL> SELECT FILE_ID, TABLESPACE_NAME, FILE_NAME, STATUS, AUTOEXTENSIBLE, MAXBYTES/
1024/1024/1024 AS "MAX_SIZE(GB)", ONLINE_STATUS FROM DBA_DATA_FILES WHERE FILE_ID=5;
```

FILE_ID	TABLESPACE_NAME	FILE_NAME	STATUS	AUT	MAX_SIZE(GB)	ONLINE_
5	SHA	+DATA/PROD/sha01.dbf	AVAILABLE	YES	31.9999847	RECOVER

5. Set New OS Destination and Restore Datafile

Use RMAN to specify the new location on the file system (/u01/app/oracle/sha01.dbf), restore, and switch the datafile.

```
RMAN> RUN {  
  SET NEWNAME FOR DATAFILE 5 TO '/u01/app/oracle/sha01.dbf';  
  RESTORE DATAFILE 5;  
  SWITCH DATAFILE ALL;  
}
```

```
executing command: SET NEWNAME
```

```
Starting restore at 07-SEP-2026 14:05:03  
using channel ORA_DISK_1
```

```
channel ORA_DISK_1: starting datafile backup set restore  
channel ORA_DISK_1: specifying datafile(s) to restore from backup set  
channel ORA_DISK_1: restoring datafile 00005 to /u01/app/oracle/sha01.dbf  
channel ORA_DISK_1: reading from backup piece +FRA/PROD/BACKUPSET/2026_09_07/  
nnndf0_tag20260907t135532_0.268.1243346133  
channel ORA_DISK_1: piece handle=+FRA/PROD/BACKUPSET/2026_09_07/  
nnndf0_tag20260907t135532_0.268.1243346133 tag=TAG20260907T135532 comment=NONE  
channel ORA_DISK_1: restored backup piece 1  
channel ORA_DISK_1: restore complete, elapsed time: 00:00:15  
Finished restore at 07-SEP-2026 14:05:18
```

```
datafile 5 switched to datafile copy  
input datafile copy RECID=2 STAMP=1243346719 file name=/u01/app/oracle/sha01.dbf
```

6. Recover the Datafile

Apply necessary archivelogs to bring the restored datafile up to date.

```
RMAN> RECOVER DATAFILE 5;
```

```
Starting recover at 07-SEP-2026 14:06:47  
using channel ORA_DISK_1
```

```
starting media recovery  
media recovery complete, elapsed time: 00:00:01
```

```
Finished recover at 07-SEP-2026 14:06:49
```

7. Bring Tablespace Online

```
SQL> alter tablespace sha online;  
Tablespace altered.
```

8. Verify New Datafile Location

Verify that the datafile has been successfully relocated to the new filesystem path and is online.

```
SQL> SELECT FILE_ID, TABLESPACE_NAME, FILE_NAME, STATUS, AUTOEXTENSIBLE, MAXBYTES/  
1024/1024/1024 AS "MAX_SIZE(GB)", ONLINE_STATUS FROM DBA_DATA_FILES WHERE FILE_ID=5;
```

FILE_ID	TABLESPACE_NAME	FILE_NAME	STATUS	AUT	MAX_SIZE(GB)	ONLINE_
5	SHA	/u01/app/oracle/sha01.dbf	AVAILABLE	YES	31.9999847	ONLINE